Systems Analysis & Design

Semester 2026-I

Workshop No. 1

Systems Analysis - Local business support recovery

Group Seven

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System Overview and Scope

System Description

The system analyzed is Titan Plaza shopping mall, located on Av. Boyacá with Calle 80, one of the most visited shopping malls in Bogotá. It was selected as a case study for the "Local Business Recovery Support System" project, which aims to develop an analytics platform to support small businesses' post-pandemic recovery through customer engagement and inventory optimization.
Although Titan Plaza houses large chains such as Falabella and Éxito, it also contains numerous small independent businesses that are the focus of this analysis. The shopping mall was chosen because it concentrates a wide variety of small businesses in one place, facilitating data collection and allowing observation of how these shops face challenges related to customer traffic, inventory turnover, and loyalty in a real world environment.
This analysis of Titan Plaza will provide real world insights into small businesses' challenges, serving as the foundation for designing the Local Business Recovery Support System in subsequent workshops.

Main Components and Boundaries

Titan Plaza has 4 floors dedicated to commerce, 250 stores, a food court, a cinema, department stores such as Falabella, and more than 2,000 parking spaces. The shopping mall is open daily from 8:00 a.m. to 10:00 p.m.
Within the scope of this analysis are the small independent businesses, their employees, customers, and common areas such as hallways, rest areas, and food courts.
Excluded from the scope are large chain stores, the mall's administrative offices, loading and unloading areas, external suppliers, and stores located in the surroundings of the shopping mall.

Stakeholders

- Customers: People who visit the mall to shop, eat, or spend time with family. They look for good prices, variety, and a pleasant experience.
- Employees: They assist customers and manage inventory daily. They know firsthand how work flows and what customers need.
- Store Owners: Entrepreneurs or managers responsible for sales, inventory, and profitability. They seek to adapt to new shopping habits.
- Mall Administrators: They manage common areas, security, and overall mall operations. Their goal is to keep the mall attractive and busy.
- Suppliers: They provide the products sold in stores. Their relationship with store owners is key to keeping inventory available.

Context

This analysis takes place in the post-pandemic context (2024-2026), where consumer behavior has significantly shifted. Studies show that customers now prioritize value for money, prefer open and ventilated spaces, combine online and physical shopping, and seek experiential retail over mere product acquisition (McKinsey, 2023).
Titan Plaza is located in northwestern Bogotá, attracting residents from middle-income neighborhoods (strata 3-5).
This analysis will identify real needs of small merchants regarding customer engagement and inventory optimization, serving as foundation for the proposed system.

Data Collection Plan and Execution

Methodology Design

- **Direct Observation:**

  Through direct observation, the flow of people entering and exiting the shopping center was analyzed, as well as the behavior of visitors within its different areas. During the observation, the areas with the highest concentration of people and the activities carried out by users in specific locations were recorded.

  From this observation, it was identified that some of the busiest areas are the cafes, especially Starbucks and Tostao, as well as the gym. This may be because many people use these spaces not only to purchase products, but also to work, study, or meet. Additionally, it was observed that Sundays tend to be days with a greater presence of families.

- **Surveys:**

  Information was collected from employees and customers who frequent the shopping center using a survey method. The questions focused on identifying the areas they visit most often within the mall and how they perceive the flow of people and revenue before and after the pandemic.

  Based on the responses, some patterns were identified regarding visitor preferences and their perception of the mall's operations. The results allowed for a comparison of how foot traffic has changed since the pandemic and which spaces or activities are currently most frequented by users.

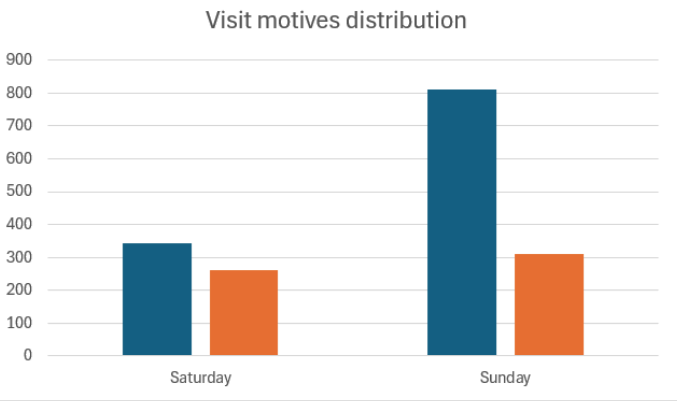
- **Environmental Correlation:**

Using the environmental correlation method, different environmental factors that could influence the flow of people within the shopping center were analyzed. For this purpose, observation sessions were conducted on two different days (Saturday and Sunday), taking into account variables such as weather, time of the month, in-store promotions, and the type of visitors present.

Primary Data Presentation

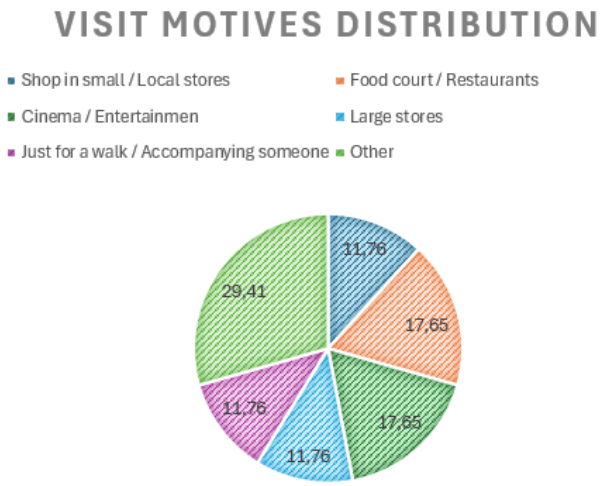
- **Graphic 1:** Foot Traffic Comparison

ANSWER	ENTER	EXIT
Saturday (9am - 12pm)	343	262
Sunday (12pm - 3pm)	811	309
TOTAL	1154	571



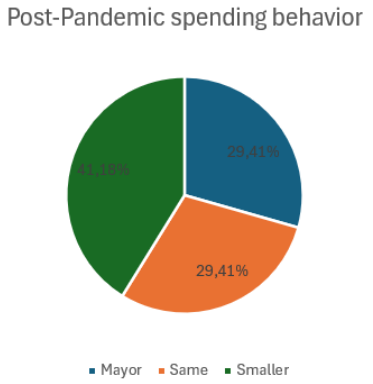
- **Graphic 2:** Visit motives distribution

ANSWER	FREQUENCY	%
Shop in small / Local stores	2	11,76%
Food court / Restaurants	3	17,65%
Cinema / Entertainmen	3	17.65%
Large stores	2	11.76%
Just for a walk / Accompanying someone	2	11.76%
Other	5	29.41%
TOTAL	17	100%



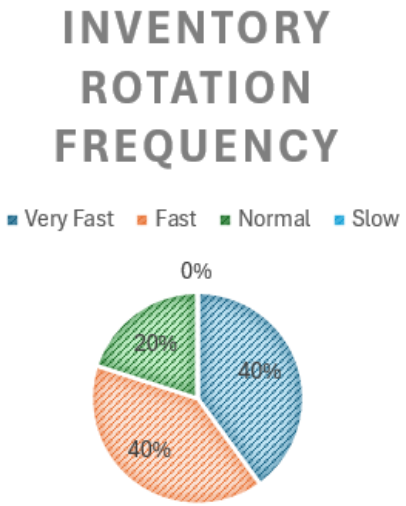
- **Graphic 3:** Post-Pandemic spending behavior

ANSWER	FREQUENCY	%
Mayor	5	29,41
Same	5	29,41
Smaller	7	41,18
TOTAL	17	100%



- **Graphic 4:** Inventory Rotation Frequency

ANSWER	FREQUENCY	%
Very Fast (less than a week)	2	40%
Fast (1 - 2 weeks)	2	40%
Normal (2 - 4 weeks)	1	20%
Slow (more than a month)	0	0%
TOTAL	5	100%



Data Quality Assessment

Although the data collected allowed us to obtain useful information about the behavior of visitors and the flow of people in the shopping center, it is important to consider some limitations of the collection process:

1. Firstly, the number of surveys carried out was relatively small, so the results represent only a sample of user behavior and do not necessarily reflect all visitors.

2. There may also be possible biases in the responses, since some people may have responded based on their personal perception or individual experiences, which may influence the objectivity of the results.

3. On the other hand, some external factors, such as the weather, promotions in stores or the specific day of the week, could influence the flow of people during the observation period.

4. Finally, some challenges arose during data collection, such as limited time to conduct observations and the availability of people willing to respond to the surveys.

Despite these limitations, the combination of several collection methods allowed obtaining an overview of the system behavior and improving the reliability of the analysis.

System Analysis

Element and Relationship Analysis

The local business recovery system is composed of several elements that interact with each other, such as local businesses, customers, business managers, online platforms, and local authorities.

Businesses offer products and services, while customers generate demand through consumption. Business managers implement strategies to adapt to market conditions, and online platforms facilitate the promotion and sale of products. Local authorities can influence the system through regulations or support measures.

The interaction between these elements creates a network of relationships in which changes in one component can affect the behavior of the system as a whole.

Sensitivity and Variability Study

The system is sensitive to changes in different factors, such as customer behavior, business strategies, economic conditions, weather, and the level of attendance at the shopping mall.

Temporal Variability (Weekday vs. Weekend): The system exhibits high variability based on the day of the week. Observing the data collected on Sundays, it shows a state shift where the customer flow changes from gym and co-working during weekdays to cinema and food Court. While foot traffic increases on weekends, the conversion rate for small businesses does not scale proportionally, as recreational users tend to prioritize entertainment over specialized local shopping. This behavior may be due to the fact that larger chain stores dominate this business area; meanwhile, small merchants are focused on the sale of products or services, rather than entertainment.

Sensitivity to Product Availability: A critical sensitivity point identified in the data is the lack of identification of customer needs to offer products accordingly. A large portion of surveyed customers reported being "Frequently" or "Sometimes" unable to find specific items in small stores. The system is highly sensitive to this gap; a single negative experience regarding

availability triggers a loop where the customer permanently redirects their demand toward large anchor stores (Falabella/Éxito) or e-commerce platforms.

Influence Sensitivity: The decision-making process of the customer is highly sensitive to two main variables: Price and Recommendations. Survey results indicate that local businesses are vulnerable to price fluctuations, for example, the cost of leasing premises within the shopping center. Furthermore, a significant sensitivity to "Word-of-Mouth" was detected, meaning the system's stability depends heavily on informal information flows rather than structured marketing.

Complexity and Emergent Behavior

The small business ecosystem in Titan Plaza functions as an adaptive system where individual interactions between customers and local merchants give rise to patterns that cannot be explained by looking at the parts in isolation

Feedback Loops and System Complexity: Reinforcing Loop (Word-of-Mouth): A positive impact with a small business triggers a word-of-mouth effect. This activates a feedback system that directly increases the sales effectiveness of small merchants, as informal recommendations stabilize their position within the complex mall environment.

Balancing Loop: Large chain stores can diminish the potential customer base of small shops. If a small store fails to offer a unique value proposition or personalized attention, the system automatically redirects the customer flow toward the more "stable" large-scale nodes.

The complexity of the ecosystem is heightened by the lack of a unified data interface, where each small shop operates as an isolated agent with limited information regarding the overall customer flow patterns detected in our analysis. This lack of connectivity leads to suboptimal inventory decisions, a key emergent property of a fragmented system that becomes evident when compared to large retail chains; the latter, by possessing direct digital contact channels, allow customers to verify availability online and go directly to the physical store, eliminating the possibility of the user exploring or discovering small businesses during their visit. Consequently, it can be inferred that if local merchants had a similar direct contact medium, they could identify the most sought-after products and dynamically adapt their inventory to actual customer needs, mitigating the current informational disconnect within the system.

Findings and Pattern Recognition

From the data collected through direct observation, surveys and environmental assessment, several paterns in the behavior of visitors to the shopping center could be identified:

Direct observation allowed us to identify:

- Some areas have a greater concentration of people, especially cafes, restaurants and the gym
- These places are not only used to consume products, but also as places to work, study or meet.
- Generating a greater permanence of visitors in these areas

The surveys carried out with clients and employees allowed us to identify:

- Trends in the way people use the shopping center, many visitors indicated that they frequent the location primarily to eat visit shops or spend free time.
- Some employes noted that the flow of people has changed before and after the pandemic suggesting a variation in visiting and consumption patterns.

Environmental Correlation allowed us to identify:

- The flow of people varies depending on the day of the week and external conditions (such as weather and so on):
  For example, Sunday saw a considerably greater flow of visitors, mainly families coming to the shopping center for lunch or a walk.
- There are also some factors such as promotions in stores or weather conditions, such as rain, that can influence the increase in the number of visitors or decrease it.

Critical Assessment

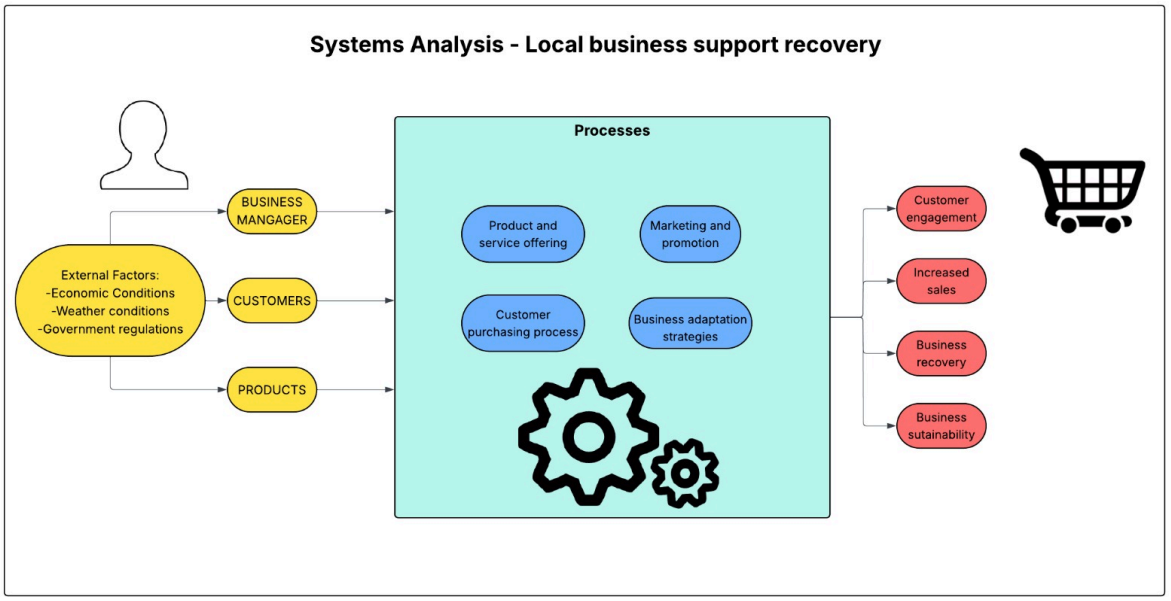
System performance evaluation
The system analysis of the small business ecosystem in Titan Plaza reveals a significant structural disconnect between potential demand and effective supply. Although the shopping center successfully attracts a massive flow of users through entertainment nodes such as gyms and cinemas, the system fails to transform this traffic into growth for small merchants due to a lack of informational integration.

Bottleneck identification:
A critical failure is the visibility gap; while large chains use digital interconnectedness to secure customer visits, small businesses operate as isolated agents with no real-time capacity to adapt their inventory to identified customer needs. This creates a fragile system where the small merchant is highly sensitive to external variables, such as leasing costs and price fluctuations, without possessing the digital tools to mitigate these risks.

Optimization opportunities and recommendations:
From a systems engineering perspective, the ecosystem requires a transition from a fragmented model to an integrated information network that allows local businesses to

compete through data-driven adaptation, rather than relying solely on accidental pedestrian flow.

Visual System Representation



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